

April 21, 2026

Synthesis of Maine Hospital Financial Data Exploration

Context

Given the imperative to ensure adequate healthcare access throughout our state and that hospitals provide a set of services to treat a unique level of acuity, the state needs to have a high confidence view on whether any of our existing hospitals are at risk of closing for financial reasons. To do this, we are intending to build a model to predict hospital closures due to financial struggles using audited financial statements. In order to facilitate the research needed to build this kind of model, we put together a data exploration environment to 1) explore the financials of individual hospitals and health systems and to put them in context [[Individual Analysis](#)] and 2) look across hospitals to identify the characteristics differentiating those that have closed from those that have not [[Cross-Sectional Analysis](#)]. This report is an attempt to synthesize these dashboards with the primary goal of pulling out the dynamics that would inform building a predictive model.

The report will only cover the dynamics observed in Maine hospitals/systems from 2005 to 2024 (the full history [available through MHDO](#)). As of 2024, there are 33 hospitals in the sample and four major health systems: all of which operate as non-profits.

In the analysis period, 3 acute care hospitals closed (Northern Light Inland, St. Andrews, and Parkview Adventist) in addition to a larger number of mergers of struggling hospitals into larger health systems with the ability to cross-subsidize and share resources. Here, we will only consider true closures as opposed to these “bailouts.” However, this is an area that will need to be explored further before building a model as our hypothesis is that much of the difference between the two outcomes is situational.

Additionally, to give more robustness to the findings in this document, the scope of the analysis should be extended to include more non-profit systems across states. We are prioritizing using the Audited Financial Statements as opposed to Medicare Cost Reports for this research given the former is better standardized for comparing across entities and the definitions for several common ratios are more helpful to the analysis. Not every state requires hospitals or systems to report audited financials but several including California, Colorado, Connecticut, Florida, Massachusetts, Oregon, and Vermont do. Currently data for Massachusetts are additionally loaded into the system. The architecture is built to make adding new states straightforward and we would need to prioritize setting up the scraping and configuration of these data.

Summary

The hospital and system financials should give us a reasonable starting point to create a coarse but useful model to assess a hospital's current risk of closing but is unlikely to give us the tools to create a finer model that predicts forward-looking changes in risk.

1. **Section 1a.** For the coarse model, the best predictor of success or failure appears to be a combination of hospital margins and system ability and willingness to cross-subsidize.
2. **Section 1b.** A system's ability to subsidize affiliates is largely dependent on its profitability. In Maine, each system's profitability is very sensitive to the performance of the largest hospital in the system.
3. **Section 1c.** Given all Maine hospitals are non-profit, willingness to intervene appears to be a function of the magnitude struggling hospital's patient revenue.
4. **Section 1d.** Looking purely at the Audited Financial Statements doesn't appear to give us the ability to understand drivers of margins which would prevent us from producing a more detailed forward-looking model.

Using a back of the envelope version of this model (i.e. taking into account hospital margins, system profitability, and hospital size), the hospital ecosystem as a whole seems to be in a precarious position with several areas of acute risk.

- **Section 2a.** The fact that some of the largest hospitals in the state are performing poorly means the entire system is operating at a heightened risk.
- **Section 2b.** Given the sketch of a model laid out above, the places I'd be most concerned about today are:
 - The big medical centers outside of Portland (Eastern Maine Medical Center and Central Maine Medical Center)
 - Northern Light AR Gould
 - York Hospital
 - St. Mary's Regional Medical Center

Details

(1a) As expected, the best predictor of success or failure appears to be a combination of hospital margins and system ability (margins) and willingness to cross-subsidize.

Methodological Framework

In attempting to identify hospitals at risk of closing, the approach we'll take in this report is to look for characteristics that separate failed hospitals as a population from those that are still operational. A few considerations:

- **Audited Financial Statements vs Medicare Cost Reports.** For this analysis, we're only looking at Audited Financial Statements vs. Medicare Cost Reports. The differences between the two are well documented online, so I won't go into them here. But for our purposes, Audited Financial Statements makes more sense as it is designed to make comparisons across entities more consistent. Its primary limitation is we'll get less granularity of revenues and costs than Medicare Cost Reports would provide.
- **Endpoint vs Moving Average.** The data we're working with tends to be quite noisy year over year. When we think shocks or big moves are important, we'll use endpoints. In cases where we think trend or performance through time matters more, we'll look at moving averages.^{1 2}
- **Normalizing variables.** Unless we explicitly want to take a hospital's size into account in a variable, we'll want to normalize our measures either by using a ratio or looking at percent changes to make comparisons across entities of different sizes comparable.
- **Beware data mining.** Given the small sample size (particularly of closures), we'll need to be cautious about drawing strong conclusions just from a relationship in the data. To increase our validity, we'll want to approach the analysis with strong priors about how we'd expect hospital closures to work vs. simply identifying the empirically strongest relationships.

¹ Two analytical notes on moving averages: (1) for moving averages of ratios, this analysis uses the moving average of the numerator divided by the moving average of the denominator. This tends to be less choppy than a moving average of the ratio itself and has a nicer intuitive interpretation. (2) in most cases we'll use arithmetic moving averages, but in cases of percent changes, we use geometric.

² Throughout the report when evaluating measures using moving averages, we by and large look at 5-year-moving averages which, if evaluating the most recent point, encompasses 2019-2024. We considered somewhere between 3 and 5 years to be a reasonable window that would smooth over one off moves but not be so long as to contain irrelevant information. We haven't found any of the findings to be sensitive to the window lengths in this range.

- **Hospitals not yet closed.** When looking at recent points, particularly when using moving averages, we don't yet know whether a hospital that is currently operational will close in the next few years. This report's analysis doesn't correct for this potential issue and is an area of needed improvement.

Hospital Financials Simple Mental Model

Hospitals' financials are quite involved with meaningful differences from other businesses. This isn't the appropriate place to give a full primer on hospital finances and given the granularity of the Audited Financial Statements, a great level of detail wouldn't be necessary to understand this analysis. That said, having a rough mental model will be helpful to interpret the results.

1. By and large, hospitals act as independent business units responsible for their own profitability.

Like typical businesses, hospitals have operating revenues (from treating patients) and operating expenses (salaries, drugs, supplies, building maintenance, etc.). On net, for the hospital to stay in business without external support, their revenues need to exceed their expenses.

2. Under certain conditions, the larger health system will transfer funding between affiliates to cover the facilities losses, also known as cross subsidization.

The conditions for whether to provide transfers are a research question addressed in subsequent sections.

3. Liquidity tends to be managed centrally at the system level.

By liquidity, we mean assets that can be easily converted into cash in order to pay expenses. The fact that liquidity is managed at the system level means that looking at common liquidity ratios such as current ratio or days cash on hand at the hospital-level may be misleading. Access to credit isn't something considered in depth in this report, but it would also mean that the credit-worthiness of the system as opposed to the hospital is important to maintain access to capital.

Observations

1. As expected, the hospitals that have closed have negative and lower sustained margins (measured using moving averages) than most of the rest of the population in the years leading up to failing. **[Figure 1 & Figure 2]**

At least using the Audited Financial Statements, the type of profit margin (total or operational) doesn't seem to matter. **[Figure 3 & Figure 4]** The decision to incorporate one vs. another flavor of profitability margin into a model would be a matter of tuning thresholds.

This said, there are a number of cases where hospitals had worse margins & didn't close. We can see that many of these cases are instances where the system stepped in and made cash transfers to either subsidize or acquire the struggling hospital **[Figure 5]**.

The last point concerning profitability is that in change terms, softening demand (operating revenues) seem to be more problematic for closures than runaway costs (operating expenses) **[Figure 6 & Figure 7]**. In fact, in two of the cases of closures, costs associated with the hospital flattened in the years up to failing likely as a result of cutting services. My intuition is that:

- Non-profit systems are less willing to step in to support a struggling hospital where members of the community have alternatives or the hospital is serving a declining population. Both scenarios would show up in softening revenues via pricing power and patient volumes.
 - Costs are easier to pass through to consumers/insurance (though less true among hospitals with weak market power/or those that rely heavily on Medicare/Medicaid payments).
2. There does seem to be lower liquidity among closed hospitals compared to non-closed hospitals **[Figure 8 & Figure 9]**, but as noted above, given liquidity tends to be managed at the system level, it is unclear whether we'd want to incorporate this into a model for non-independent hospitals.

That is, it is unclear whether low hospital liquidity for affiliated hospitals is an accounting trick or has a real-world meaning. It remains an open question how to interpret this phenomenon.

3. [Chartis logistic regression](#) uses average age of plant but seems unrelated for ME sample **[Figure 10]**
4. No other measures jumped out in particular as being substantially different between operational hospitals and ones that have closed.

(1b) The ability of a health system to be a financial support to struggling hospitals is very sensitive to the profit margins of its largest hospital.

A system will only be able to subsidize/stabilize its struggling affiliates by:

1. Generating and reallocating positive profits
2. Spending down its balance sheet
3. Receiving gifts/government subsidies

Given a system will not be able to sustain spending down its assets indefinitely and one time government payments seem to be minor part of the financial picture outside of extraordinary times (e.g. Covid), the primary mechanism for a system being able to support its affiliates is through its profits.

One interesting feature about the Maine health systems is that each system has a single hospital that accounts for over 50% of revenues. Given its size, we observe that this largest hospital consequently dominates whether the system as a whole is profitable.

For example, MaineHealth has Maine Medical Center (7.7% 5-year average operating margin) which over the last five years has averaged \$1.9B in revenues out of the systems' \$3.6B total revenues. Maine Medical Center also contributes almost all of their historic outperformance **[Figure 11 & Figure 12]**. The system was able to achieve profitable margins despite only 7 out of 11 of the hospitals in the system being on average profitable over that period.

Northern Light on the other hand (-3.1% 5 year avg. operating margins) **[Figure 13]** has Eastern Maine Medical Center as its largest member which had modestly negative margins (-2.2%) but again accounts for over half of system revenues **[Figure 14]** and contributed > 65% of the system losses. **[Figure 15]** This being the case, the system as a whole is unprofitable despite a higher percentage of the hospitals being profitable than MaineHealth. One unfortunate corollary to this is that Northern Light Inland's 2024 closure won't meaningfully change the system's overall profitability.

This dynamic implies an unintuitive but potentially useful policy lever which is for non-profit systems, one way to improve the stability of rural or smaller hospitals with consistent demand but low performance is to improve the margins of large urban hospitals with higher revenues. That is, some MaineHealth hospitals such as Maine Behavioral Health and Franklin Memorial have performed consistently at levels that would have forced restructuring or closure if they belonged to a worse performing system or were independent **[Figure 16 & Figure 17]** but are able to continue operating largely as a result of transfers from Maine Medical Center. **[Figure 18]** This implication appears to hold for the sample, but before endorsing this from a policy perspective, we would want to test the robustness of the hypothesis on a larger sample of hospitals from other states.

(1c) Given all hospitals in this sample are non-profit, willingness to intervene appears to be a function of a struggling hospital's volume.

Even in cases where a system is healthy, we see cases where a system will allow an affiliate to fail (St. Andrews) or no system will step up to acquire the struggling independent hospital (Parkview Adventist). This leads to the question: in which cases will a healthy system step in to subsidize or acquire a struggling hospital vs. not?

Our working hypothesis is the system's decision comes down to whether the struggling hospital is still serving a vital purpose or not. In a first order sense, this can be approximated with volume (preferably utilization, but within our dataset, revenue is our best proxy). We expect but have not tested that other considerations such as distance to the next facility providing comparable services would be important as well.

Looking at our case studies, St. Andrews' owner Lincoln Health Care (a subsidiary of MaineHealth) [cited low occupancy](#) as a primary reason for not saving St. Andrews' hospital despite it hovering around -3% 5-year moving average margins in 2013. **[Figure 19]** On the other hand, Franklin Memorial (also a MaineHealth affiliate) was allowed to reach -9.4% 5-year moving average margins at its low point **[Figure 20]**. We suspect the decision to save Franklin Memorial is largely due to the fact it's serving a much larger population **[Figure 21]**

(1d) Looking purely at the Audited Financial Statements doesn't appear to give us the ability to understand drivers of margins which would prevent us from producing a finer forward-looking model.

Seeing as our view is that historical margins are going to be the most important factor for determining whether a hospital requires intervention, to increase the predictive power of a model, we would need to be able to predict changes in margins. That is, looking at historical margins will be helpful to understand what the risk to a hospital is today (with a several month lag), but it would be more helpful to have a view on which hospitals will have more vs less margin pressure in the future.

As I would expect, taking a cursory look, none of line items or ratios in change or level form variables are predictive of changes in margins (ie. no r^2 values above 0.2). **[Figure 22]** I would expect you'd be able to do somewhat better with the financials than a one-variable regression but would also expect there is a ceiling working with this data. For this analysis, we're going to leave the question of how to predict changes in margins out of scope, and just note some questions that came up through the rest of the exercise.

Given the prior discussion, it seems necessary to understand the drivers of diverging performance for the largest hospitals in the state. Maine Medical Center has a level-one

trauma center and serves a larger population than the other large hospitals, but it would be worth getting tighter on what characteristics are driving the divergence in performance.

Additionally, there are several rural hospitals that have seen explosive margin growth despite it being a generally difficult environment for hospitals. Several cases including Calais, Down East, Northern Light Acadia, Northern Light Mayo, and Rumford Hospital [Figure 23, Figure 24, Figure 25, Figure 26, Figure 27] would be worth exploring further.

If one were to pick up this work and attempt to create such a model, some of the variables up front I would consider include: payer mix, geography, distance from other facilities, acuity of care, service mix, and wages by job classification. Much of this data seems to be available in the National Academy of State Health Policy's dataset of Medicare Cost Reports, so it seems tractable to at least take a swing in producing a competent model.

(2a) The fact that some of the largest hospitals in the state are performing poorly means the entire system is operating at a heightened risk.

As noted in Section 1b, the ability for a healthcare system to be a financial support is very sensitive to the financial stability of the largest hospital in the system. Outside of MaineHealth's Maine Medical Center, the largest hospitals in the state are performing poorly. Both Central Maine Medical Center and Northern Light Eastern Maine Medical Center are meaningfully unprofitable. [Figure 28 & Figure 29]

Given these large hospitals account for such a large portion of the systems' revenues, to the extent that other hospitals in this system become unprofitable, they will not be able to expect financial support from their parents. At present, this does not seem to be an acute issue outside of Northern Light AR Gould which in normal times would likely be a candidate for a cross-system cash transfer given its size, but the Northern Light Health system is less able to today due to Eastern Maine Medical Center's poor financial performance. [Figure 30]

An additional knock-on effect to this situation is that since fewer systems are profitable, there are fewer candidates that would be able to absorb a struggling independent hospital. Currently, MaineHealth is the only system capable for such a move.

(2b) Using a back of the envelope of this model, there are a handful of hospitals that I would be most concerned about.

As mentioned previously, this report was written prior to encoding or formally testing a model. That said, we can take our high-level observations of what properties are important and use that framework to identify the riskiest hospitals.

To formalize this a bit, the different criteria I'm using are:

1. Does a hospital have sustained negative operating margins using (5-year moving average)?
2. Does the hospital's system appear to be willing and able to intervene?
 - a. Ability: are the systems' profits sufficiently greater than the hospitals losses?
 - b. Willingness: is the hospital large? I.e. > ~\$30mm in revenues.

This is obviously an imperfect/hand-wavy methodology, but directionally aligned with the important factors we've observed. Running our data through these criteria, the hospitals with greatest risk today are:

1. The big medical centers outside of Portland (CMMC & NLH EMMC)

These are unprofitable and their systems' smaller affiliates are unable to prop up their losses. That said, there are no closures of large hospitals this data sample, so it is not entirely clear whether there are additional criteria (e.g. government intervention) that may influence their ability to stay open.

2. Northern Light AR Gould

As mentioned above, AR Gould is relatively large and modestly unprofitable. If the broader Northern Light Health system was profitable, they might expect cross-subsidies, but as it stands, the system appears unable to do so consistently.

3. York Hospital is the other hospital in an unsustainable state. **[Figure 31]**

York is currently not part of a system and given its size (~\$200mm Net Patient Service Revenue) would be a candidate for a merger. MaineHealth is the only profitable system left and at \$94mm net income, they would be able to absorb York's \$8mm annual losses. It is outside the scope of this report to predict whether a merger would in fact go through.

4. St. Mary's Regional Medical Center.

St. Mary's is an affiliate of Covenant Health, a non-Maine owned system, with substantial sustained losses (several recent years of > \$30mm) **[Figure 32]**. Given that Covenant isn't part of a Maine system, we don't have the same level of detail into their financial status more broadly. However, if Covenant was generating positive profits, our model would predict that given its size, it would be eligible for transfers which we haven't seen in the data **[Figure 33]**.

Charts

Charts 1a.

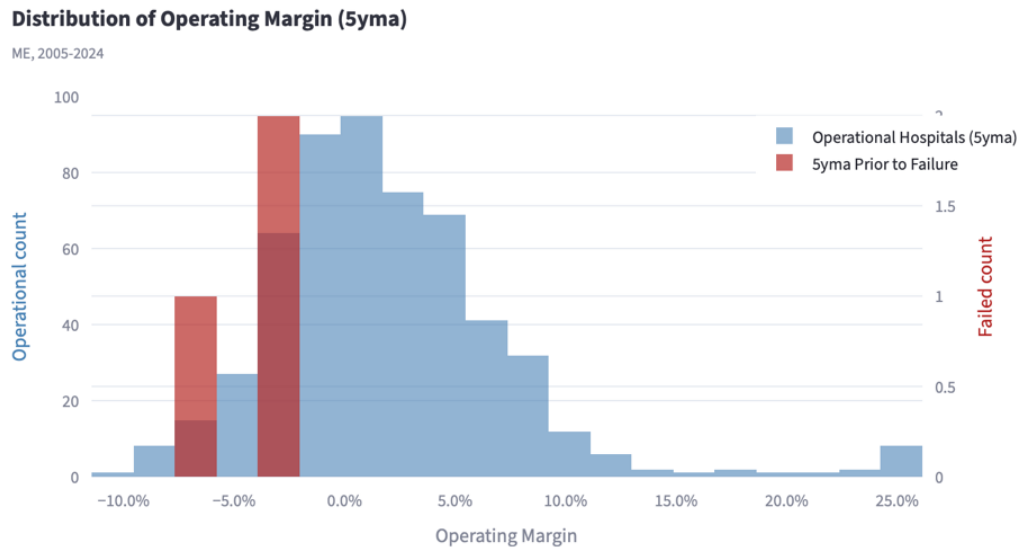


Figure 1: Distribution of all observations of 5-year moving average operating margins of Maine hospitals within the sample. The blue population includes all observations for hospitals that haven't closed. The red contains one observation per hospital that did close from years T-6 to T-1.

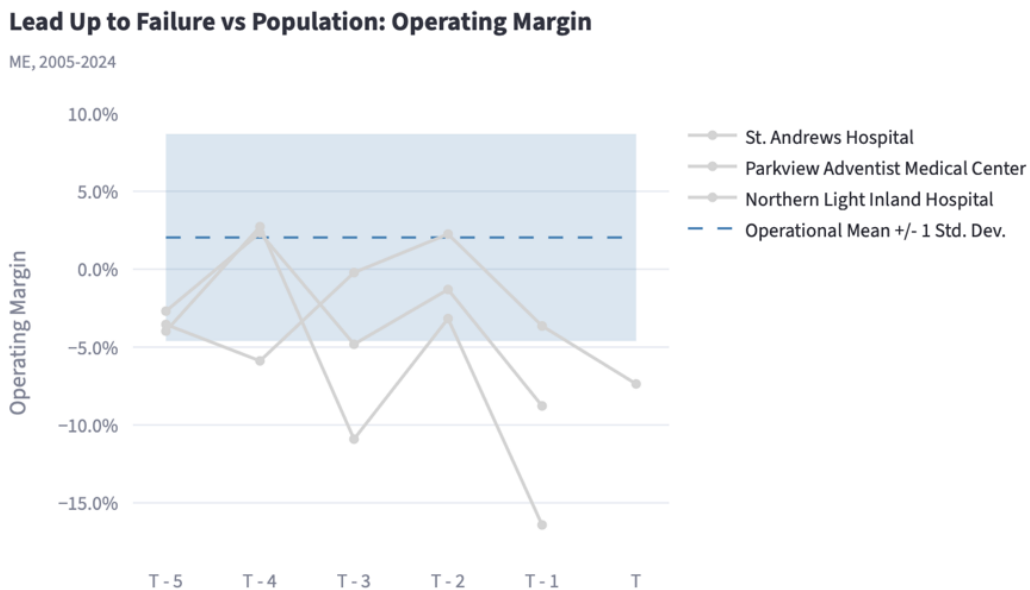


Figure 2: Path of margins of hospitals that closed in the years prior to failure vs. the mean +/- 1 std. deviation of the blue distribution above.

Distribution of Total Margin (5yma)

ME, 2005-2024

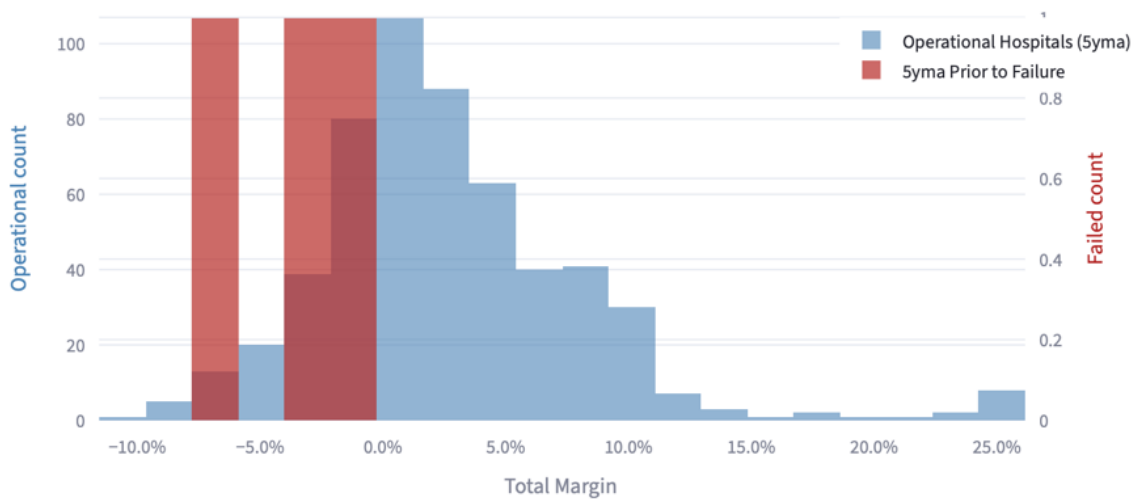


Figure 3: Distribution of total margins looks similar to operating margins in Figure 1.

Operating Margin vs Total Margin

ME, 2005-2024

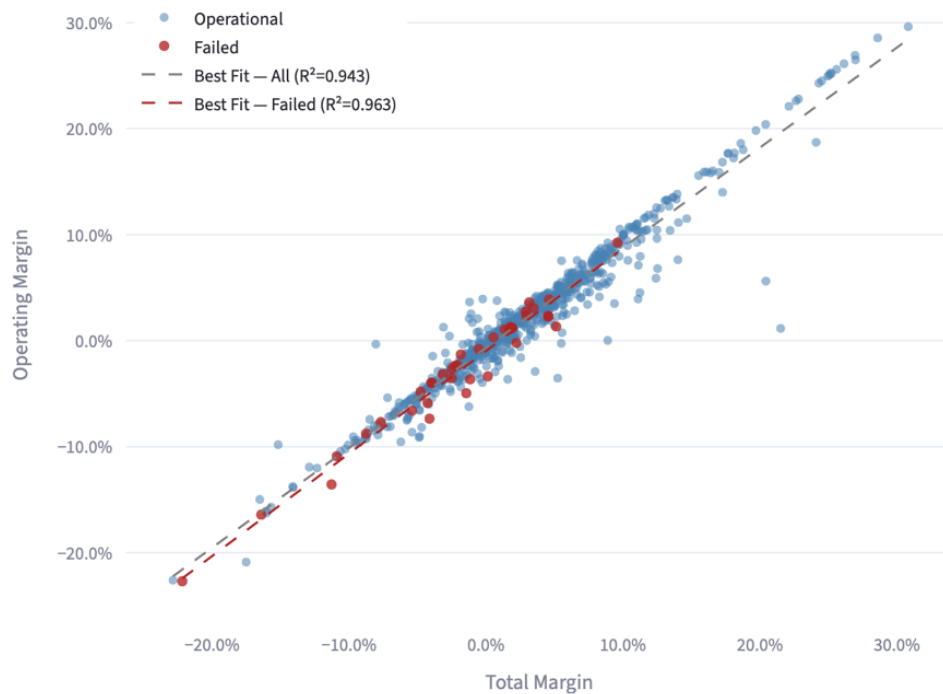


Figure 4: Avg Total Margins and Avg Operating Margins are tightly correlated.

Transfers from/to Affiliates vs Net Income(3yma)

ME, 2005-2024

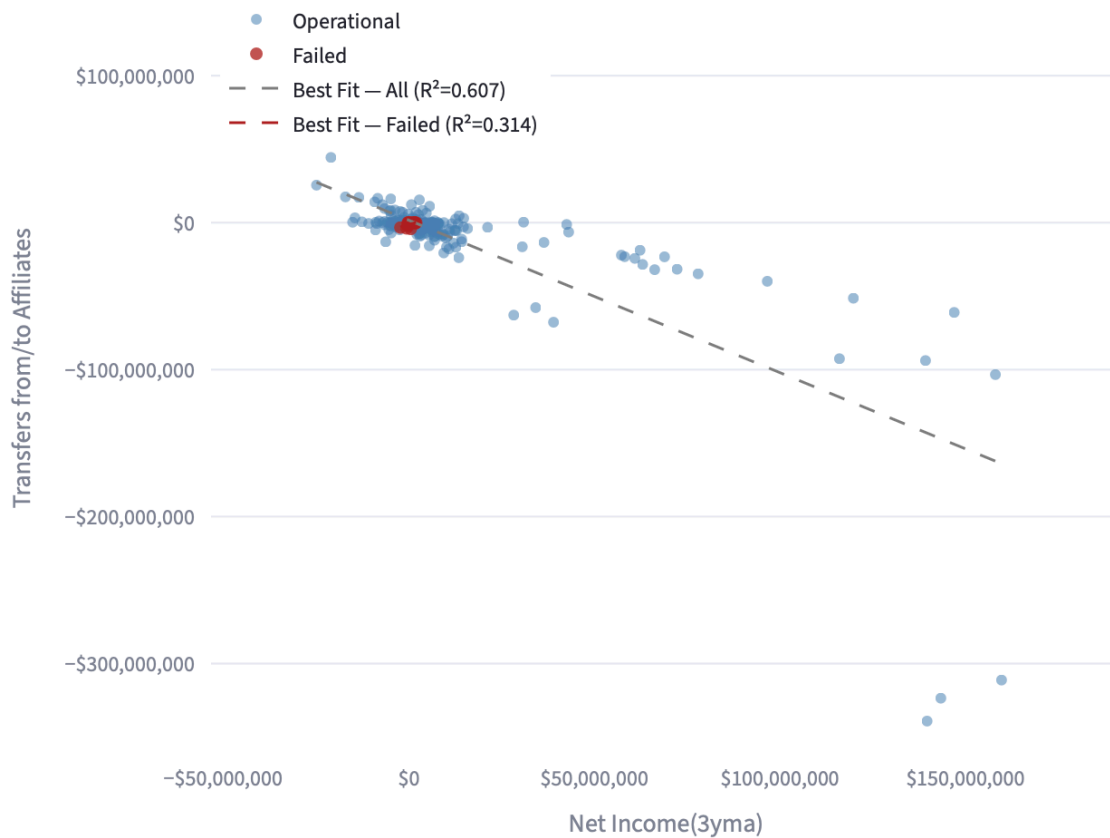


Figure 5: Transfers from/to Affiliates plotted against Net Income. In general, we observe the relationship we expect which is that hospitals with an operating loss (negative net income) receive more transfers while, the hospitals with larger net incomes fund those transfers. There are a handful of points with negative net incomes and 0 transfers. Most of these are St. Mary's Regional Medical center which is part of Covenant Health group which has historically been unprofitable. Northern Light Mercy makes up most of the remainder but in years prior to their acquisition by Northern Light.

Lead Up to Failure vs Population: % Change in Total Operating Expenses

ME, 2005-2024

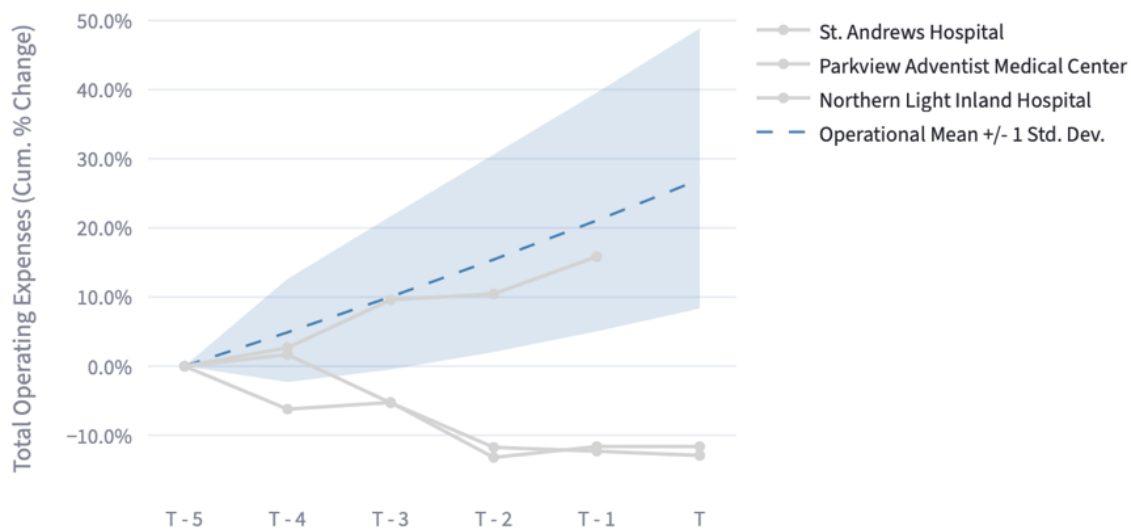


Figure 6: Operating Expenses prior to failure. In none of the cases are we seeing excessive cost increases prior to failure.

Lead Up to Failure vs Population: % Change in Total Operating Revenue

ME, 2005-2024

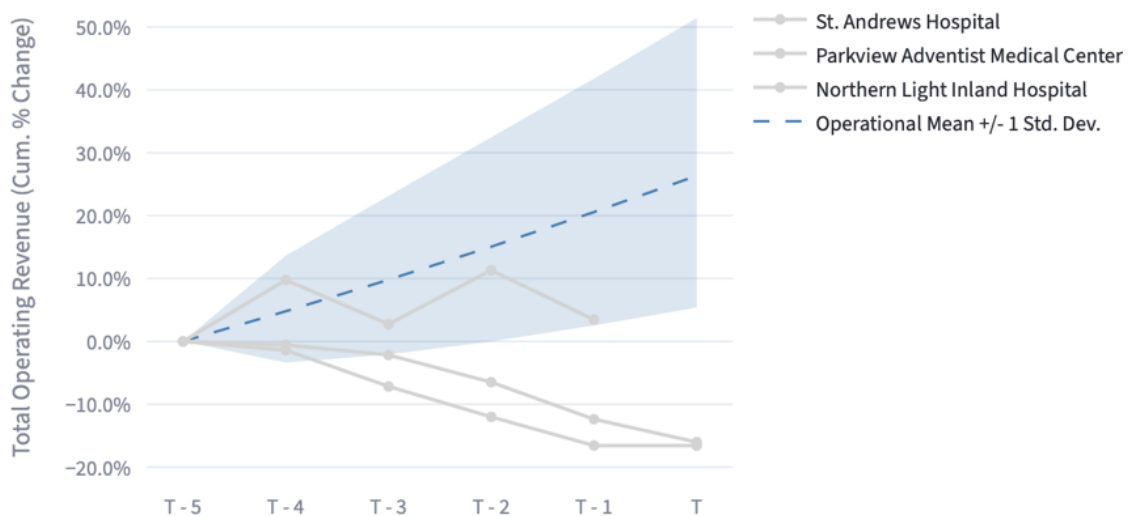


Figure 7: Operating Margins prior to failure. Below average or even decreasing revenues are apparent in all cases.

Lead Up to Failure vs Population: Current Ratio

ME, 2005-2024

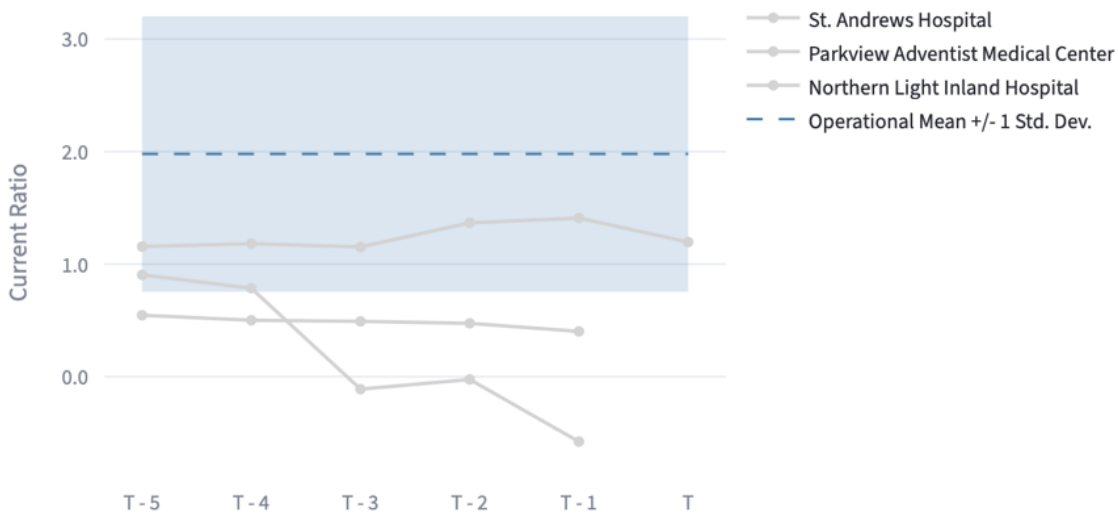


Figure 8: Liquidity measures of failed hospitals vs non-failed population. There is clearly a difference, but I'm not sure whether I'd want to include this as part of a model at the hospital level for non-independent hospitals given they manage liquidity centrally.

Lead Up to Failure vs Population: Current Days Cash on Hand

ME, 2005-2024

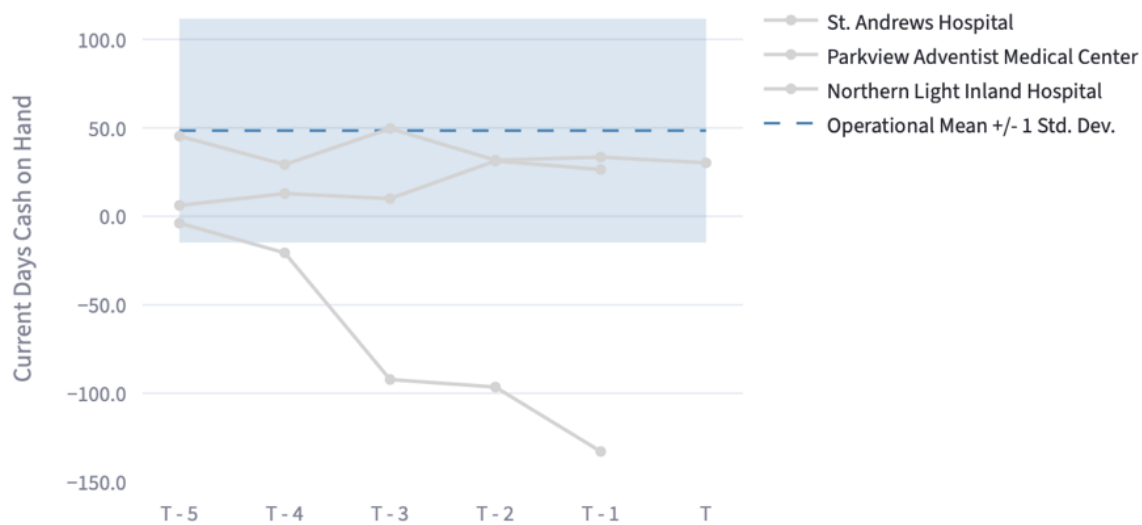


Figure 9: Liquidity measures of failed hospitals vs non-failed population. There is clearly a difference, but I'm not sure whether I'd want to include this as part of a model at the hospital level for non-independent hospitals given they manage liquidity centrally.

Distribution of Average Age of Plant (5yma)

ME, 2005-2024

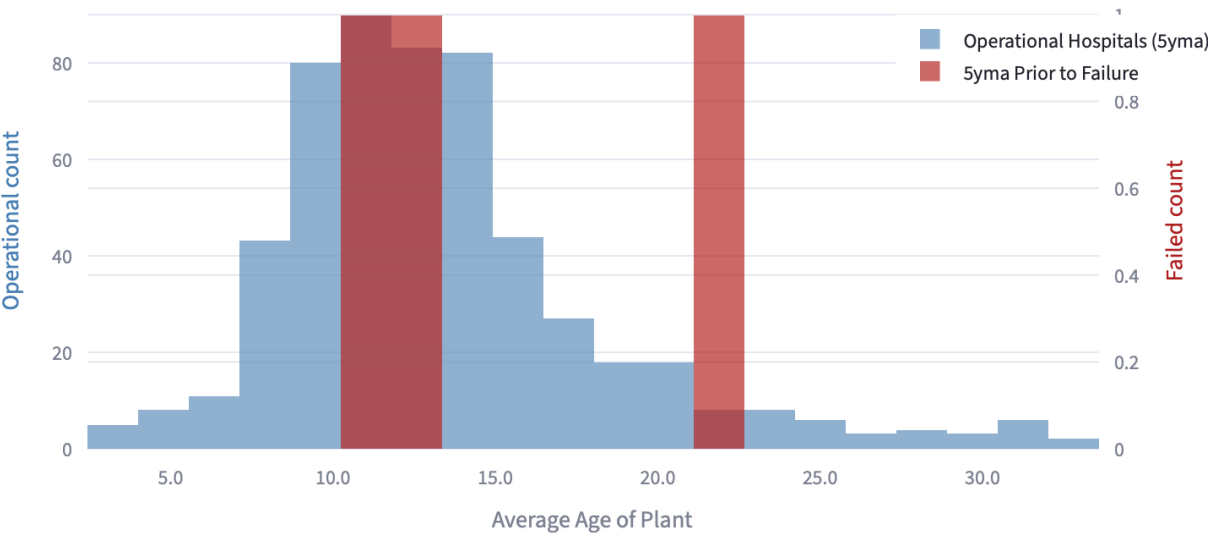


Figure 10: Northern Lights Inlands' facility was quite old before failing, but not in the extreme of the tail and the other two hospitals are squarely in the mode of the distribution.

Charts 1b.

Organization	State	Net Income (2024)	↓ Net Income, 5yma
Maine Medical Center	ME	\$214,988,208	\$147,114,052
Stephens Memorial Hospital	ME	\$12,582,035	\$12,080,118
Waldo County General Hospital	ME	\$12,693,706	\$11,950,344
LincolnHealth	ME	\$9,902,859	\$6,642,358
Southern Maine Health Care	ME	\$13,590,723	\$-798,214
Pen Bay Medical Center	ME	\$-7,881,598	\$-1,004,025
St. Joseph Hospital	ME	\$3,605,000	\$-1,312,200
Mid Coast Hospital	ME	\$2,018,006	\$-3,358,539
Franklin Memorial Hospital	ME	\$-7,204,339	\$-4,520,935
Maine Behavioral Health	ME	\$-16,698,144	\$-11,968,318
St. Mary's Regional Medical Center	ME	\$-4,766,000	\$-20,832,787

Figure 11: 2024 and 5-year moving average (ending 2024) Net Income for MaineHealth Hospitals. Only 4 out of 11 of the hospitals have been profitable and Maine Medical Center is able to cover all of the losses.

Operating Margin (5yma): MaineHealth System

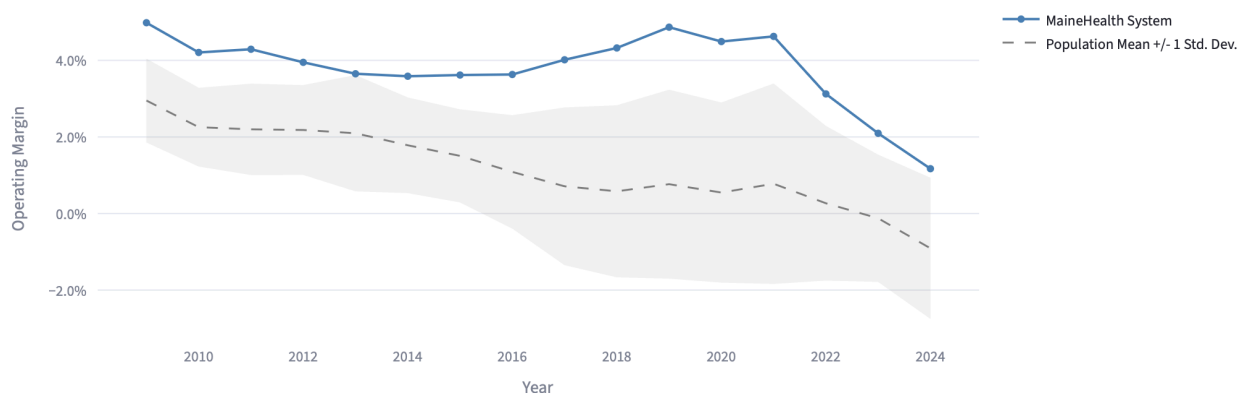


Figure 12: MaineHealth Operating Margins (5-year moving average). They have significantly outperformed peer systems and independent hospitals.

Operating Margin (5yma): Northern Light Health

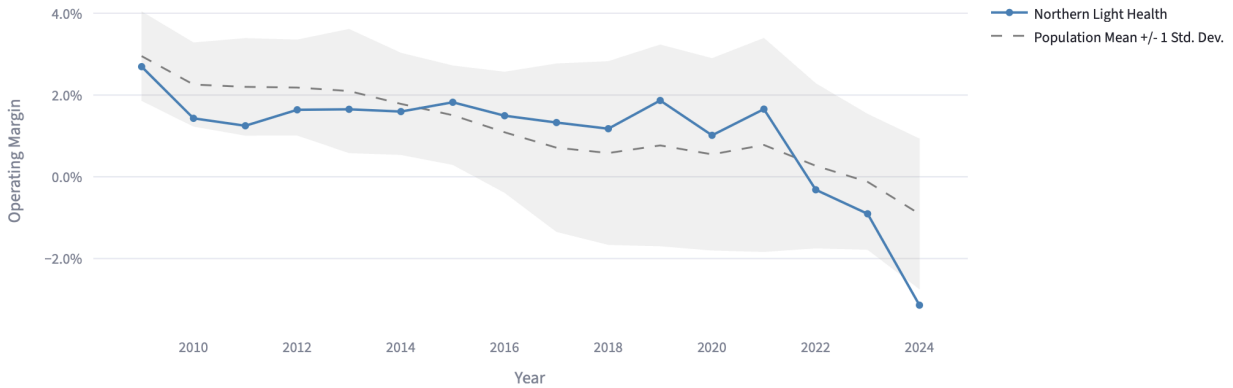


Figure 13: 5-year moving average Northern Health Operating Margins. System was consistently profitable until 2019 and has been on a sharp decline since then.

Organization	Total Revenue (2024)	↓ Total Revenue, 5yma
Northern Light Eastern Maine Medical Center	\$1,021,360,723	\$1,003,098,919
Northern Light Mercy Hospital	\$325,908,829	\$265,965,030
Northern Light AR Gould Hospital	\$168,388,777	\$160,157,947
Northern Light Maine Coast Hospital	\$116,297,207	\$102,748,689
Northern Light Inland Hospital	\$76,365,643	\$77,869,881
Northern Light Acadia Hospital	\$87,575,056	\$71,196,235
Northern Light Mayo Hospital	\$73,655,767	\$64,342,351
Northern Light Sebecook Valley Hospital	\$62,284,908	\$57,887,523
Northern Light Blue Hill Hospital	\$43,466,870	\$43,967,611
Northern Light C.A. Dean Hospital	\$22,730,077	\$22,542,245

Figure 14: Revenue by hospital for Northern Light Health System. Eastern Maine Medical Center accounts for > 50% of system revenues.

Organization	Net Income (2024)	↓ Net Income, 5yma
Northern Light Seabasticook Valley Hospital	\$5,196,861	\$7,318,472
Northern Light Acadia Hospital	\$11,531,230	\$6,113,438
Northern Light Mayo Hospital	\$8,122,253	\$4,476,868
Northern Light Blue Hill Hospital	\$2,381,440	\$4,052,475
Northern Light C.A. Dean Hospital	\$-1,354,108	\$1,102,675
Northern Light Mercy Hospital	\$10,752,495	\$611,825
Northern Light Maine Coast Hospital	\$-1,788,530	\$30,695
Northern Light AR Gould Hospital	\$-9,546,613	\$-3,529,570
Northern Light Inland Hospital	\$-12,538,934	\$-4,798,232
Northern Light Eastern Maine Medical Center	\$-87,684,650	\$-26,865,738

Figure 15: 5-year moving average Net Income for Northern Light hospitals. Northern Light's financials are dominated by Eastern Maine Medical center (>50% of system revenues – see next figure) which is a net drag on the system (also >50% of the losses). This being the case, the system as a whole is unprofitable despite a higher percentage of the hospitals being profitable than MaineHealth.

Operating Margin (5yma): Maine Behavioral Health

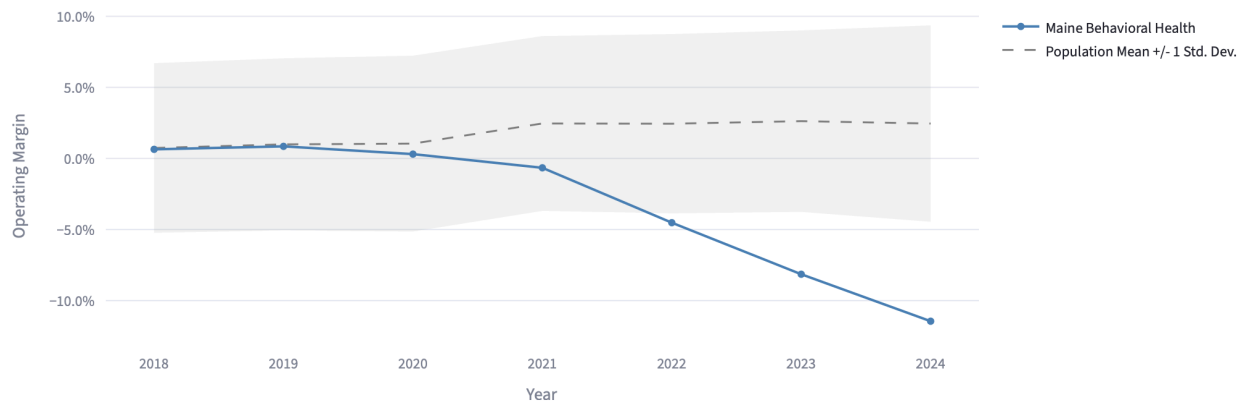


Figure 16: 5-year moving average Operating margins for Maine Behavioral Health relative to all Maine Hospitals. It has a 5-year moving average margin at its trough lower than any of the hospitals allowed to fail.

Operating Margin (5yma): Franklin Memorial Hospital

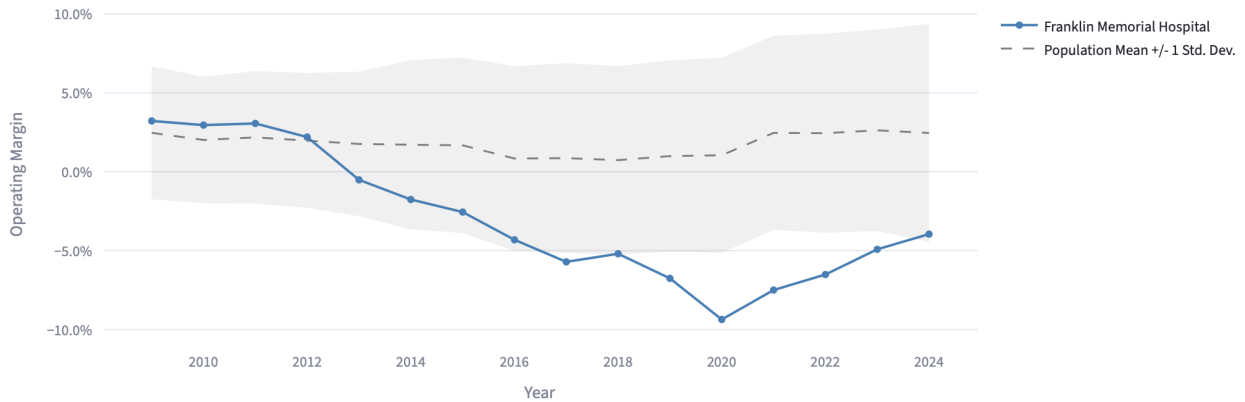


Figure 17: 5-year moving average Operating margins for Franklin Memorial relative to all Maine Hospitals. It has a 5-year moving average margin at its trough lower than any of the hospitals allowed to fail.

Transfers from/to Affiliates (Normalized): Maine Behavioral Health

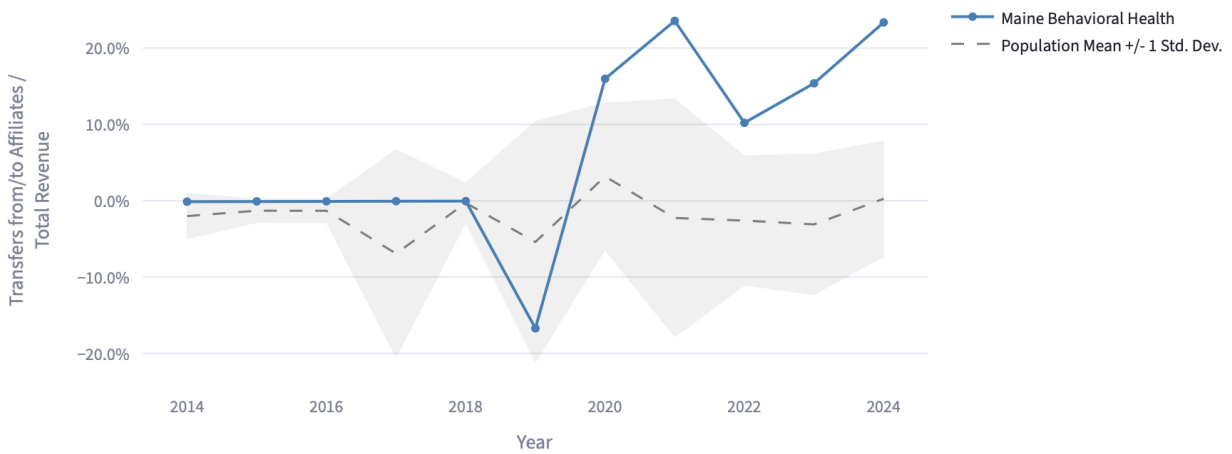


Figure 18: Maine Behavioral Health Transfers from Affiliates. Transfers tend to be on the same order of magnitude to the entity's losses.

Charts 1c.

Operating Margin (5yma): St. Andrews Hospital

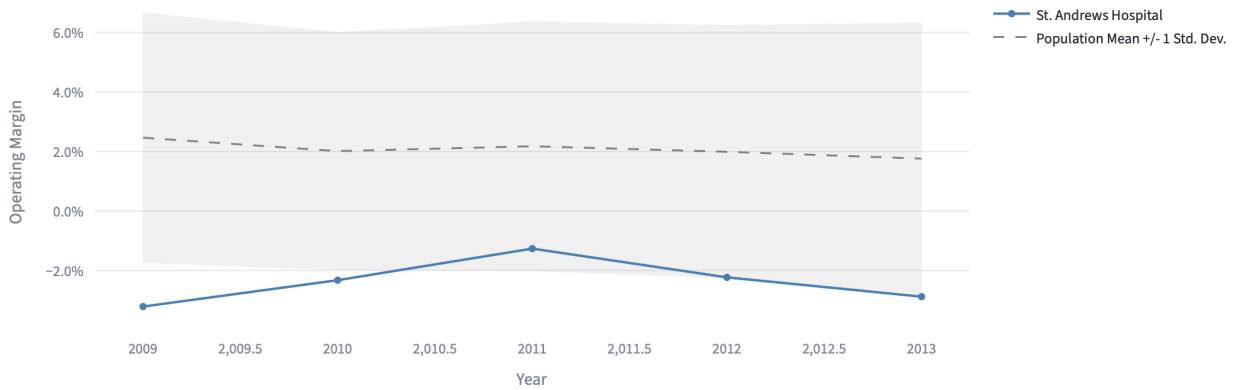


Figure 19: St. Andrews 5-year moving average operating margins.

Operating Margin (5yma): Franklin Memorial Hospital

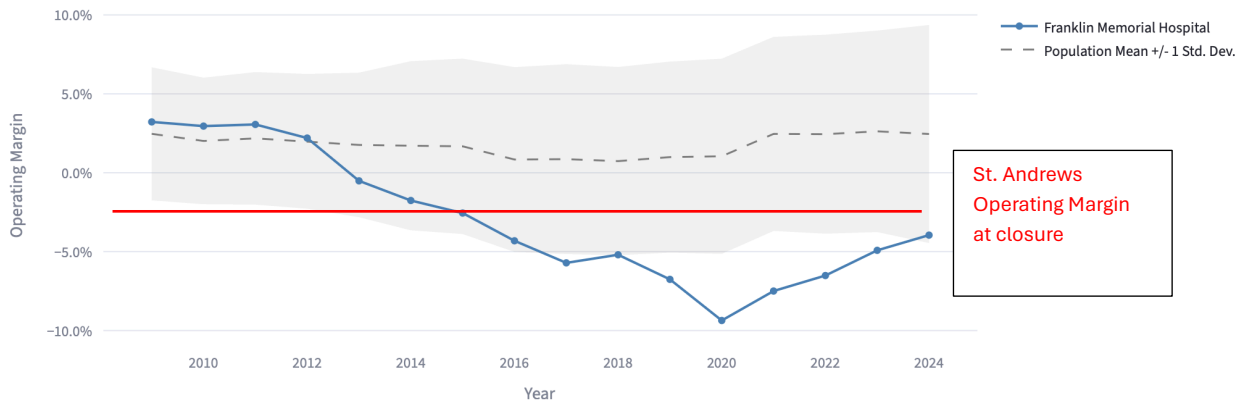


Figure 20: Franklin Memorial 5-year moving average operating margins. Annotated on the chart is the point at which St. Andrews was allowed to fail.

Organization	State	Total Revenue (2012)	Total Revenue, 5yma
Maine Medical Center	ME	\$896,981,000	\$763,536,637
Southern Maine Medical Center	ME	\$169,441,839	\$142,790,569
St. Mary's Regional Medical Center	ME	\$147,437,000	\$135,867,800
Mid Coast Hospital	ME	\$119,889,378	\$108,411,221
Pen Bay Medical Center	ME	\$117,121,918	\$113,620,605
Franklin Memorial Hospital	ME	\$89,440,874	\$82,725,375
Henrietta D. Goodall Hospital	ME	\$83,006,084	\$73,687,142
Waldo County General Hospital	ME	\$75,500,804	\$66,378,492
Miles Memorial Hospital	ME	\$57,317,489	\$52,927,976
Stephens Memorial Hospital	ME	\$56,233,885	\$46,915,379
Spring Harbor Hospital	ME	\$32,046,948	\$32,091,261
St. Andrews Hospital	ME	\$18,043,265	\$19,607,703

Figure 21: Total Operating Revenues of MaineHealth hospitals in 2012. Franklin was serving much larger demand than St. Andrews was.

R ² vs Ratios		
Measure (lag -1y)	Last R ²	MA R ²
Operating Margin	0.14	0.16
Total Margin	0.14	0.15
Return on Assets	0.13	0.14
Equity Financing Ratio	0.11	0.01
Current Ratio	0.02	0.00
Current Days Cash on Hand	0.01	0.02
Return on Equity	0.00	0.01
Average Age of Plant	0.00	0.01
Debt Service Coverage	None	None

R ² vs Ratios		
Measure (lag -1y)	Last R ²	MA R ²
Total Margin	0.14	0.00
Operating Margin	0.14	0.00
Return on Assets	0.12	0.00
Equity Financing Ratio	0.02	0.03
Current Ratio	0.02	0.03
Return on Equity	0.00	0.00
Average Age of Plant	0.00	0.00
Current Days Cash on Hand	0.00	0.00
Debt Service Coverage	0.00	0.01

R² vs Change in Income Statement Items		
Measure (lag -1y)	Last R²	MA R²
Permanently Impaired Security Write Downs	0.37	None
Net Income	0.11	0.06
Net Patient Service Revenue	0.09	0.01
Other Non-Operating Revenues (Gifts, Bequests, etc.)	0.09	0.03
Total Operating Revenue	0.08	0.01
Permanently Impaired Writedowns of Other Assets	0.04	None
Realized Gains/Losses on Sales of Securities	0.04	0.03
Net Operating Income	0.03	0.03
Gross Inpatient Service Revenue	0.03	0.03
Total Gross Patient Service Revenue	0.02	0.03

R ² vs Change in Income Statement Items		
Measure (lag -1y)	Last R ²	MA R ²
Net Operating Income	0.03	0.00
Net Income	0.03	0.00
Permanently Impaired Writedowns of Other Assets	0.01	0.06
Changes in Prior Year Estimated/Final Settlements	0.01	0.00
Permanently Impaired Security Write Downs	0.00	0.87
Bad Debt (Provision for Bad Debt)	0.00	0.03
Allowed Advertising Expenses	0.00	0.05
Total Non-Operating Revenue	0.00	0.00
Advertising Expenses	0.00	0.01
Non-Allowed Advertising Expenses	0.00	0.01

R² vs Change in Balance Sheet Items		
Measure (lag -1y)	Last R²	MA R²
Due to Affiliates (Non-Current)	0.15	0.02
Total Equity	0.04	0.01
Net Patient Accounts Receivable	0.03	0.06
Total Liabilities	0.02	0.00
Trustee Held Investments	0.02	0.00
Total Current Liabilities	0.02	0.01
Self Insurance Fund	0.02	0.04
Other Non-Current Liabilities	0.02	0.01
Other Accounts Receivable	0.01	0.00
Total Non-Current Liabilities	0.01	0.00

R ² vs Change in Balance Sheet Items		
Measure (lag -1y)	Last R ²	MA R ²
Other Non-Current Liabilities	0.01	0.00
Due to Affiliates (Current)	0.01	0.00
Due to Affiliates (Non-Current)	0.01	0.04
Due from Affiliates (Non-Current)	0.00	0.15
Third Party Settlements Receivable	0.00	0.02
Due from Affiliates (Current)	0.00	0.02
Estimated Third Party Settlements (Non-Current)	0.00	0.00
Other Accounts Receivable	0.00	0.01
Total Equity	0.00	0.00
Net Patient Accounts Receivable	0.00	0.01

Figure 22: R² of changes (left) and levels (right) of measures vs changes in operating margins sorted from greatest to least. None of these measures has much explanatory power. Permanently Impaired Security Write Downs has the highest but there's only a few values of that variable so I take that relationship to likely be spurious.

Charts 1d.

Operating Margin (5yma): Calais Regional Hospital

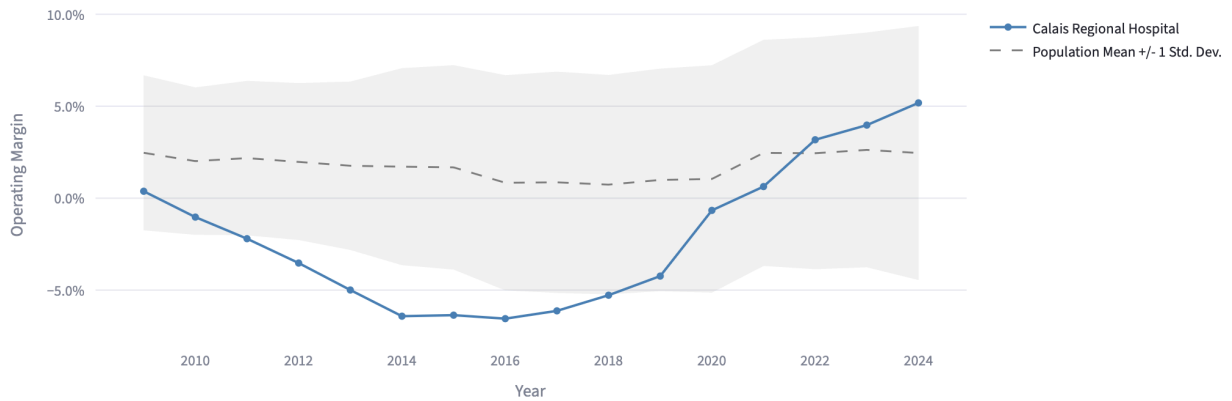


Figure 23: Calais and Downeast Community 5-year moving average operating margins have both been on a steep upward recovery since 2018 (after Calais filed for bankruptcy).

Operating Margin (5yma): Down East Community Hospital

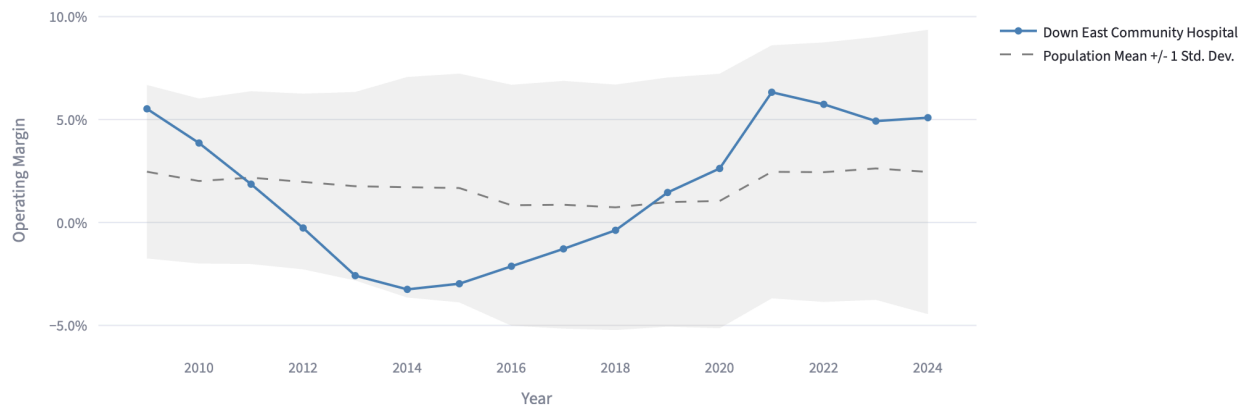


Figure 24: Calais and Downeast Community 5-year moving average operating margins have both been on a steep upward recovery since 2018 (after Calais filed for bankruptcy).

Operating Margin (5yma): Northern Light Mayo Hospital

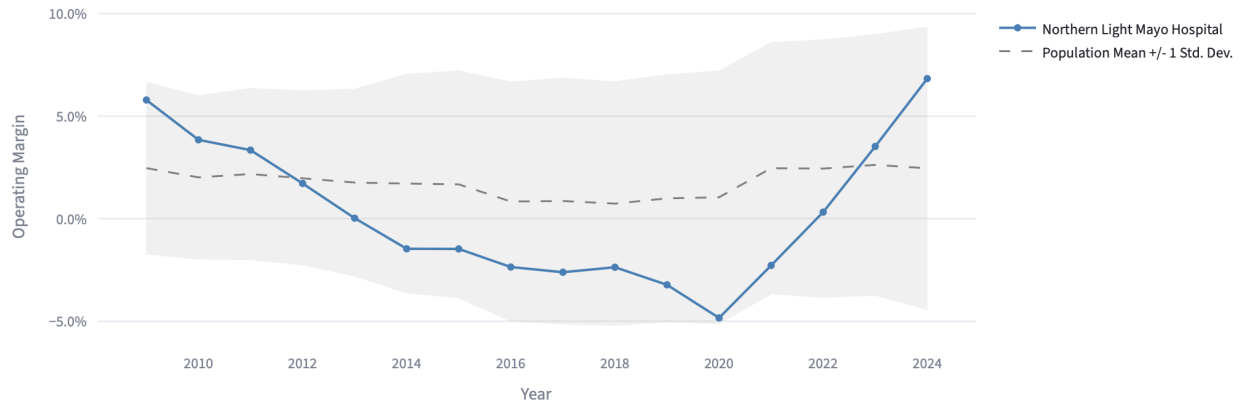


Figure 25: Northern Light Mayo and Acadia 5-year moving average operating margins.

Operating Margin (5yma): Northern Light Acadia Hospital

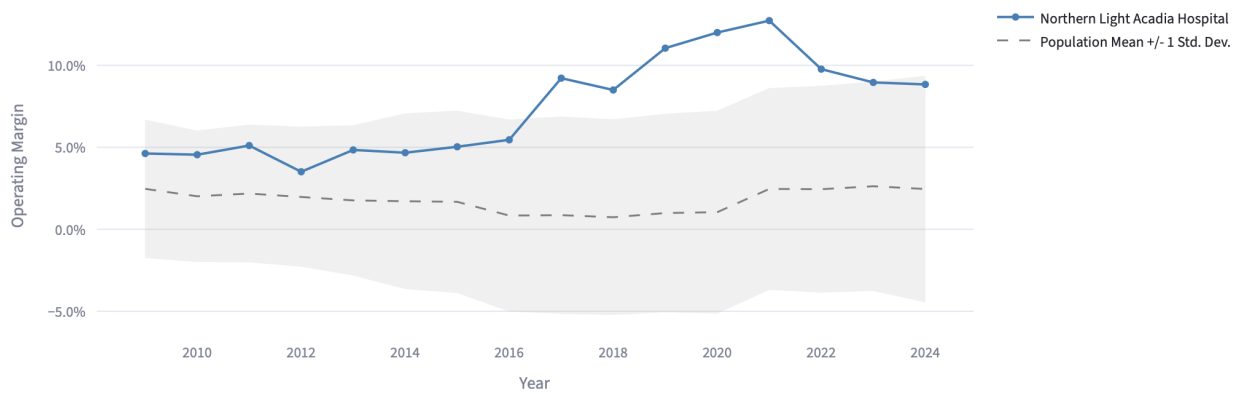


Figure 26: Northern Light Mayo and Acadia 5-year moving average operating margins.

Operating Margin (5yma): Rumford Hospital

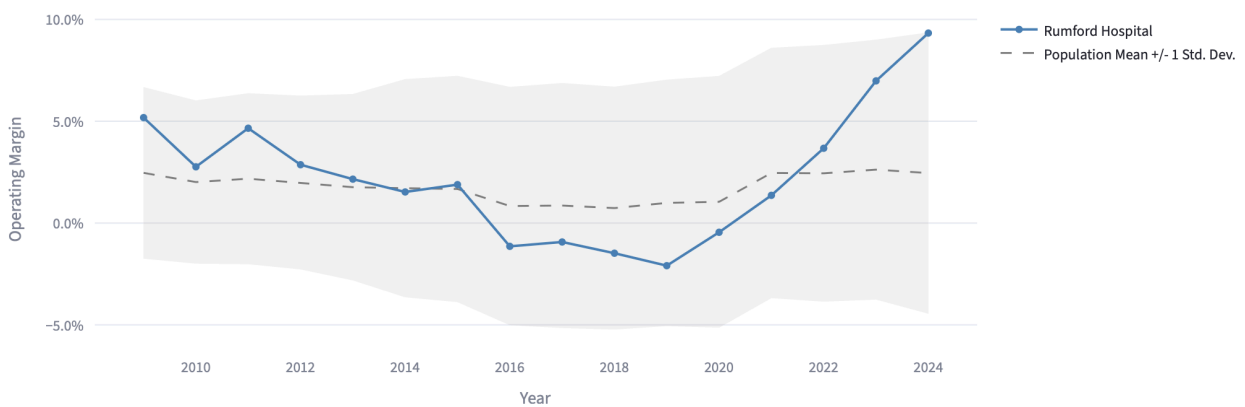


Figure 27: Rumford Hospital 5-year moving average operating margins.

Charts 2a

Operating Margin (5yma): Central Maine Medical Center

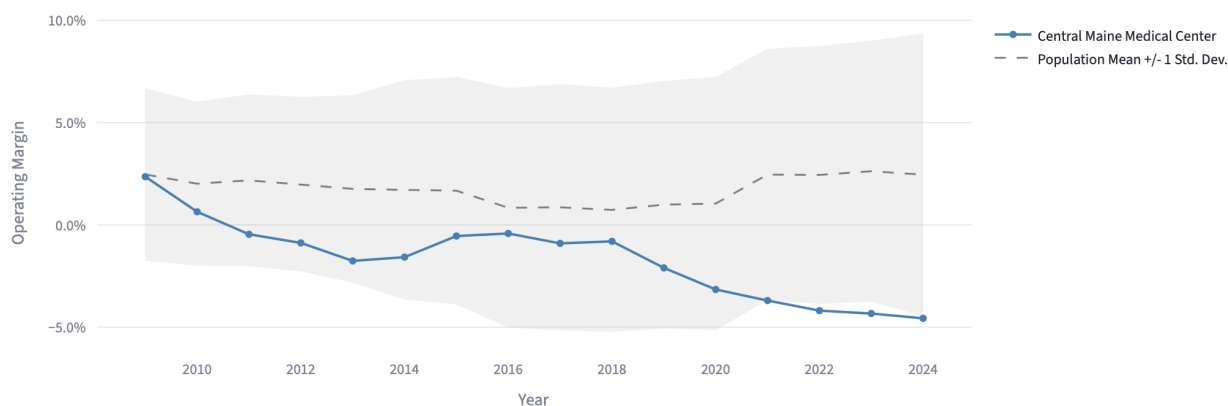


Figure 28: 5-year moving average Operating Margins for Central Maine Medical Center. It is meaningfully unprofitable.

Operating Margin (5yma): Northern Light Eastern Maine Medical Center

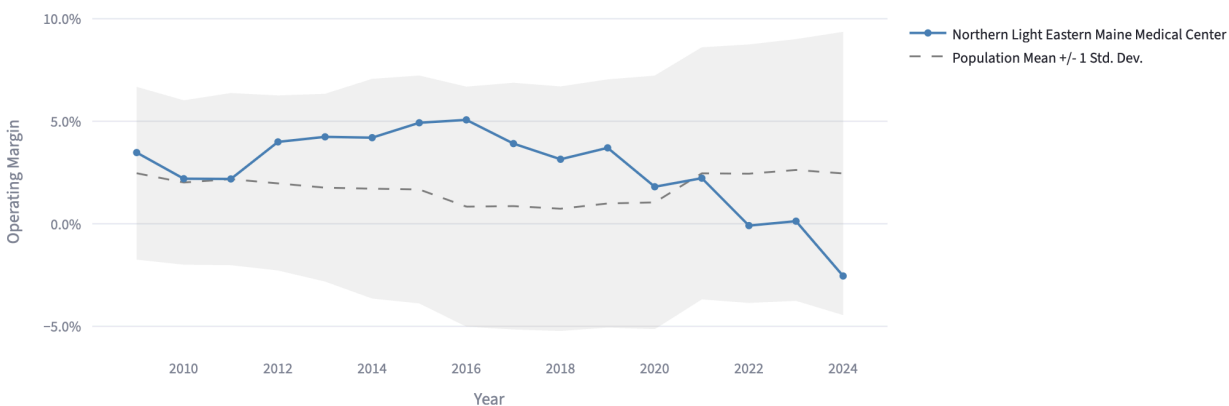


Figure 29: 5-year moving average Operating Margins for Northern Light Eastern Maine Medical Center. It has had mixed netting to poor performance over the last 5 years.

Operating Margin (5yma): Northern Light AR Gould Hospital

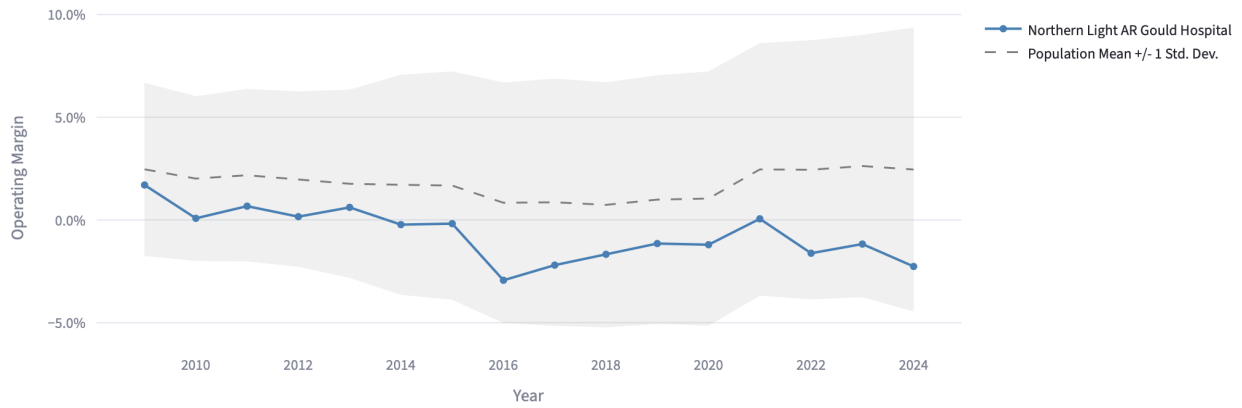


Figure 30: 5-year moving average Operating Margins Northern Light AR Gould: modestly negative.

Charts 2b

Operating Margin (5yma): York Hospital

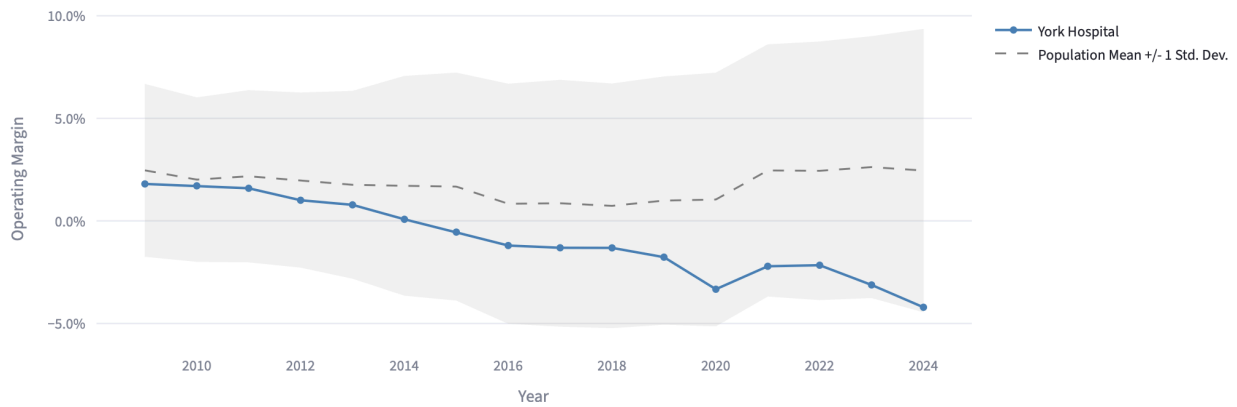


Figure 31: 5-year moving average Operating Margins York Hospital

Net Income (\$): St. Mary's Regional Medical Center

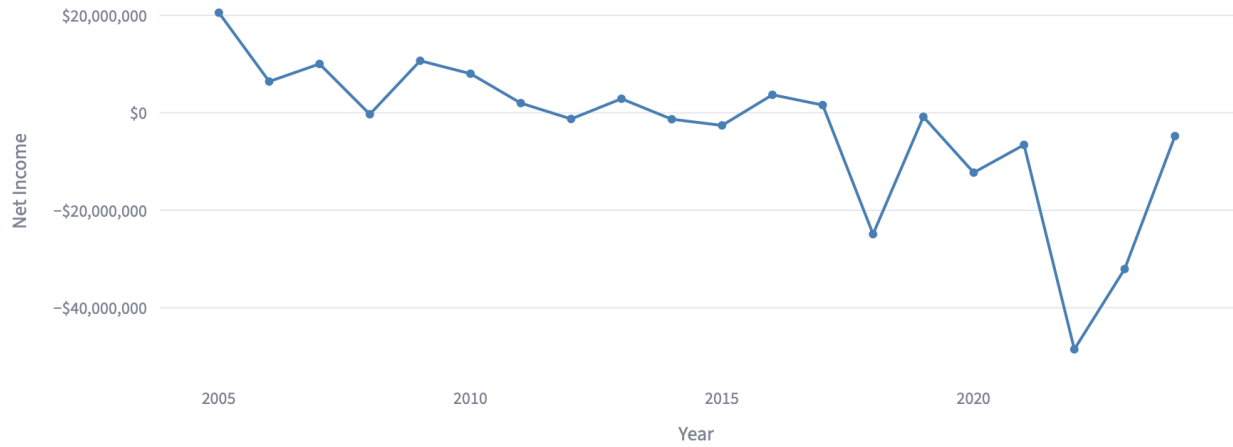


Figure 32: St. Mary's Regional Medical Center Net Income. They've sustained several recent years of substantial losses.

Transfers from/to Affiliates (\$): St. Mary's Regional Medical Center

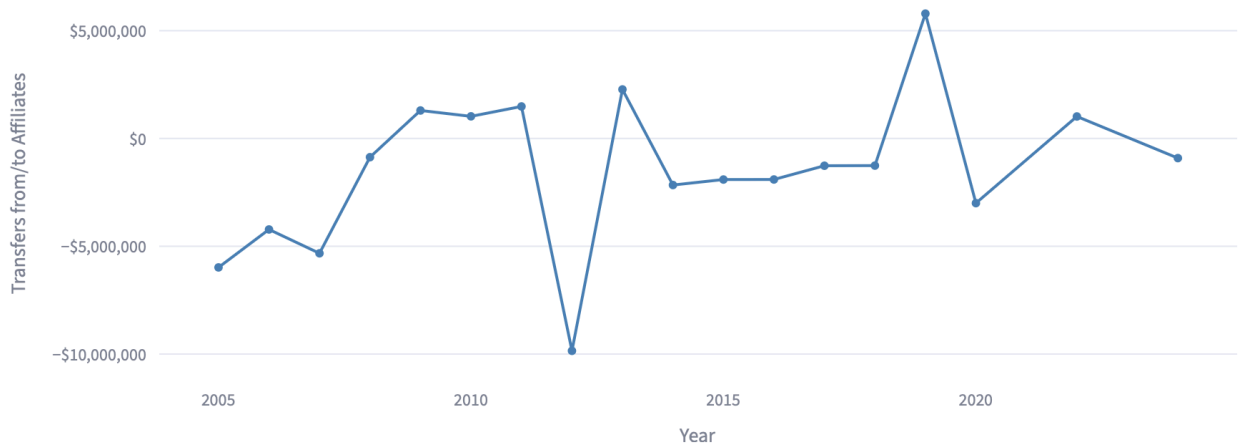


Figure 33: St. Mary's Regional Medical Center Transfers from/to Affiliates. Unlike other large struggling MaineHealth affiliates, St. Mary's hasn't received substantial subsidies.